



CoRIM Profile for Arm Confidential Computing Architecture Arm CCA Endorsements

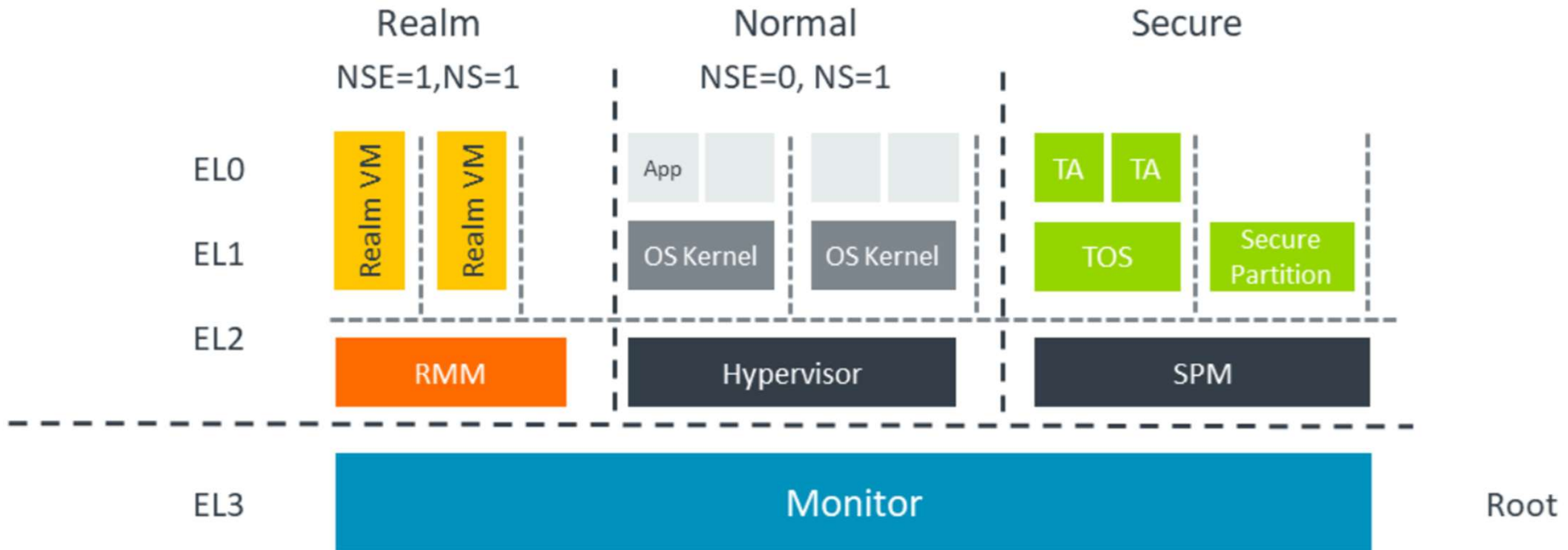
<https://datatracker.ietf.org/doc/draft-ydb-rats-cca-endorsements/04/>

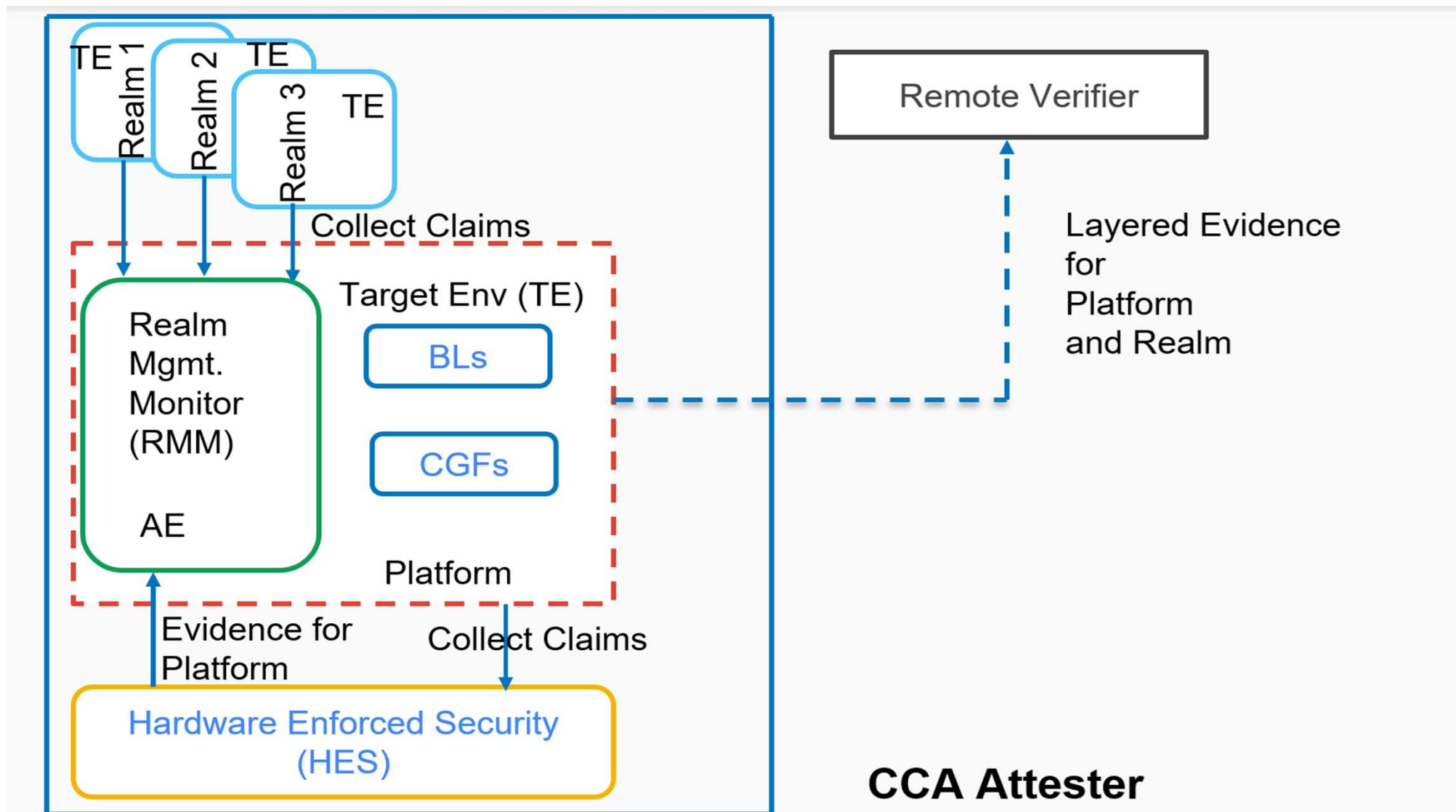
Authors – Yogesh Deshpande (Arm)

– Thomas Fossati (NVIDIA)

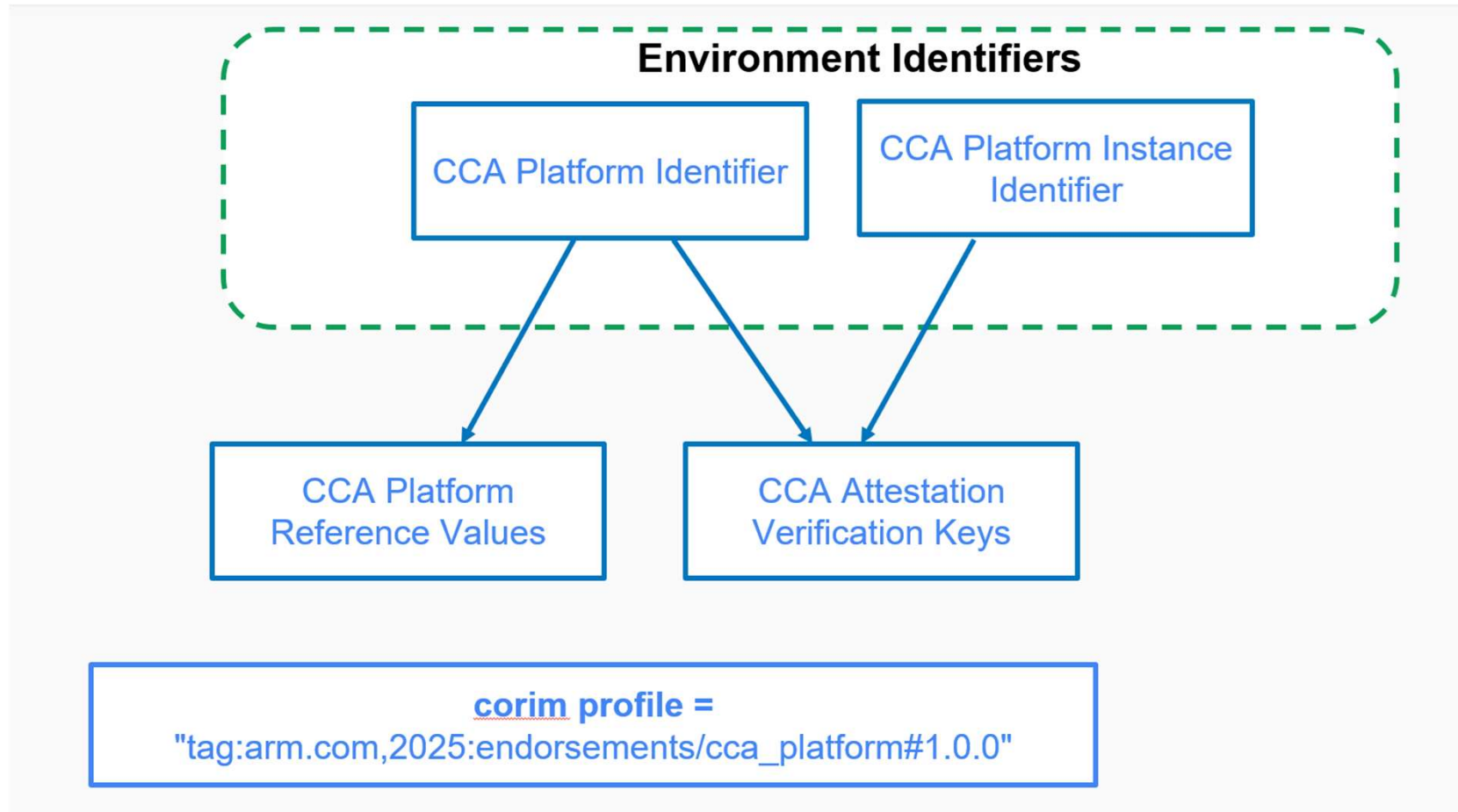
Date: 22nd September 2026

Arm CCA





Arm CCA Platform Endorsements



Platform Identifiers

CCA Platform Implementation ID

- Uniquely identifies a given implementation of a CCA platform
- Used by an Endorser or a Reference Value provider as an Anchor
- To Link Reference Values and Attestation Verification Keys for a CCA Platform
- It is a Class Identifier to identify a given CCA Platform

Encoding a CCA Platform Implementation ID – using CoRIM Environment Class

- `$class-id-type-choice = tagged-bytes` is used
- The length of the byte string MUST be exactly 32
- `impl-id-tagged-bytes = #6.560(arm-platform-implementation-id-type)`
- `arm-platform-implementation-id-type = bytes .size 32`

Platform Identifiers

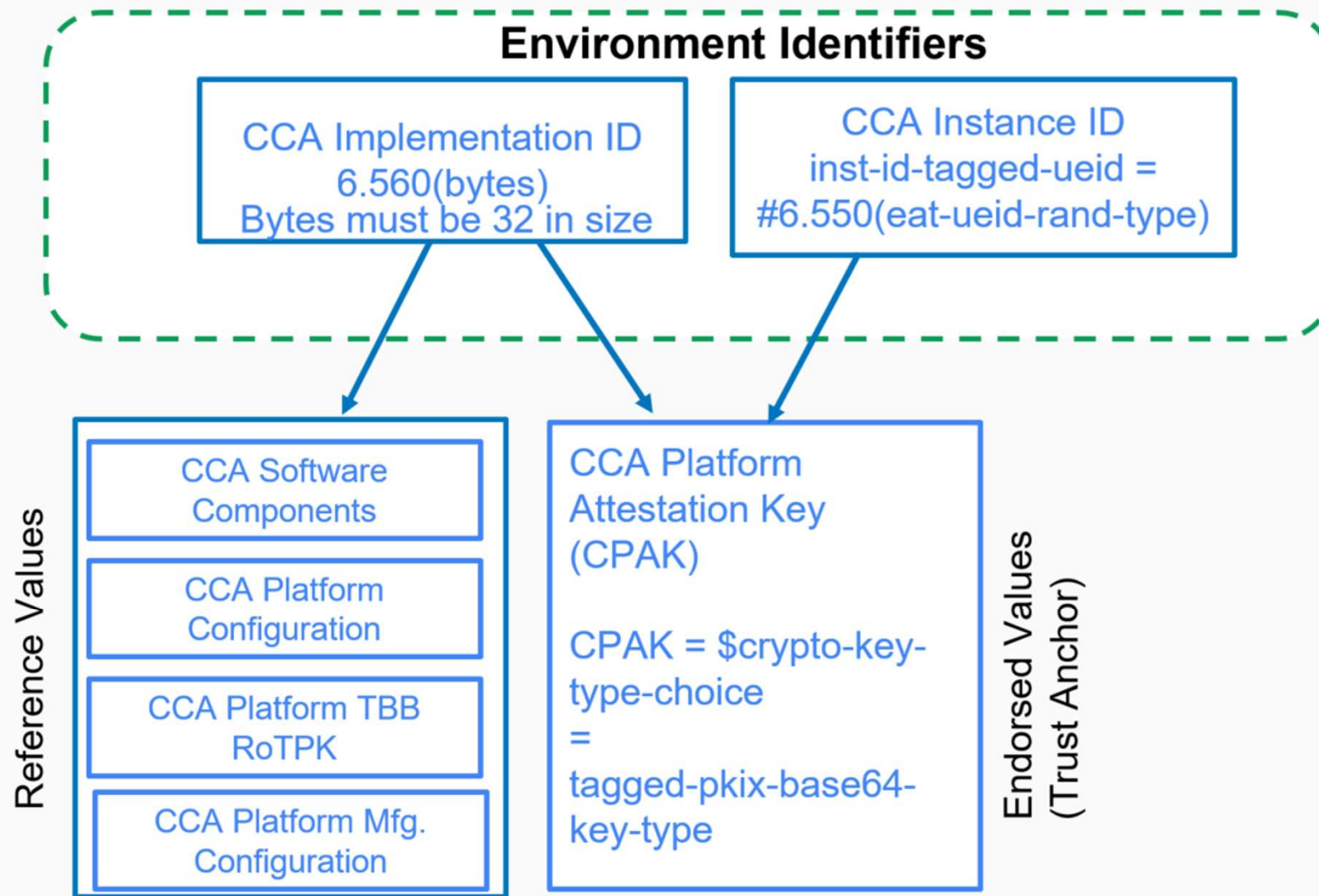
CCA Platform Instance ID

- Uniquely identifies a *specific instance* of certain CCA platform implementation
- Used by a Verifier to locate an Endorsement for the Attestation Key (PAK)

Encoding a CCA Platform Instance ID – using CoRIM Environment Instance

- \$instance-id-type-choice = tagged-ueid-type
- inst-id-tagged-ueid = #6.550(eat-ueid-rand-type)
- The first byte MUST be 0x01 (RAND) followed by 32 bytes of unique instance identifier

Environment Linkage



CCA Platform RV – CCA Platform SW Components

- List of Software Components that can affect the behavior of the CCA Platform
- CoRIM Profile contains expected Software Component Values

```
cca-swcomp-measurement-map = {  
  &(mkey: 0) => "cca.software-component"  
  &(mval: 1) => cca-swcomp-measurement-values-map  
}
```

← Mkey identifies the type of measurements

Arm CCA CoRIM Profile

cca-swcomp-measurement-values-map

? &(version: 0) => cca-swcomp-version-map

&(digests: 2) => cca-swcomp-digests-type

? &(name: 11) => cca-swcomp-name

&(cryptokeys: 13) => [cca-swcomp-signer-id]

CoRIM Draft

measurement-values-map

? &(version: 0) => version-map

? &(digests: 2) => digests-type

? &(name: 11) => text

? &(cryptokeys: 13) => [+ \$crypto-key-type-choice]

Only Component digest of size 32/48/128 is supported

Signer-id is of type “tagged-bytes”, only ONE signer ID is permitted

CCA Platform RV - CCA Platform Configuration

Set of Chosen Implementation Options of CCA platform

- For example, contains system properties
- A profile MUST have only one Platform Config Reference Value

Encoding a CCA Platform Configuration – using CoRIM “*tagged-mask-raw-value*”

```
cca-config-measurement-map = {  
  &(mkey: 0) => "cca.platform-config"  
  &(mval: 1) => cca-config-measurement-values-map  
}
```

```
cca-config-measurement-values-map = {  
  &(raw-value: 4) => cca-tagged-masked-raw-value  
}
```

```
cca-config-tagged-masked-raw-value = #6.563([  
  value: bytes  
  mask: bytes])
```

Trusted Board Boot (TBB) Root of Trust Public Keys (RoTPKs)

A List of Provisioned Public Key identifiers in a platform

- To establish a Chain of Trust
- This claim in Reference Value is OPTIONAL

Encoding a CCA ROTPKs – using CoRIM “*measurement-map*”

```
cca-rotpk-measurement-map = {  
  &(mkey: 0) => text .regexp "cca.rotpk.[CD]M\\. [0-7]\\. [0-5]"  
  &(mval: 1) => cca-rotpk-measurement-values-map  
}
```

Mkey = “cca.rotpk” +
LifeCycle State +
Active Array Index +
Position in the Array



```
cca-rotpk-measurement-values-map = {  
  &(cryptokeys: 13) => [ cca-rotpk-id ]  
}
```

```
cca-rotpk-id = #6.560(cca-hash-type)
```

```
cca-hash-type = bytes .size 32 / bytes .size 48 /  
bytes .size 64
```

CCA Platform RV - CCA Platform Manufacturing Configuration

Record of Production Phases & testing conducted during manufacturing process

- Vendor specific variable length data
- An implementation may choose to decide the value that is contained in this field

Encoding a CCA Platform Manufacturing Config

– using CoRIM “*tagged-mask-raw-value*”

```
cca-rotpk-measurement-map = {  
  &(mkey: 0) => "cca.platform-manufacturing-config"  
  &(mval: 1) => cca-manufacturing-config-measurement-values-map  
}
```

```
cca-manufacturing-config-measurement-values-map = {  
  &(raw-value: 4) => cca-tagged-masked-raw-value  
}
```

```
cca-manufacturing-config-tagged-masked-raw-value  
= #6.563([  
  value: bytes  
  mask: bytes])
```

CCA Endorsed Values – CCA Platform Trust Anchor

- Attestation Verification Key that contains the public key associated with the CCA Platform Attestation key (CPAK)
- In a Verifier, used to Verify the Integrity of the CCA Platform Attestation Token
- Associated with each Instance of a CCA Platform
- Expressed using an *attest-key-triple-record*
 - *The subject of the triple is the Environment – Identified by Implementation ID and the Instance ID*
 - *The object is the CPAK Public Key expressed using \$crypto-key-type-choice = tagged-pkix-base64-key-type*

CCA Endorsed Values – CCA Platform Trust Anchor

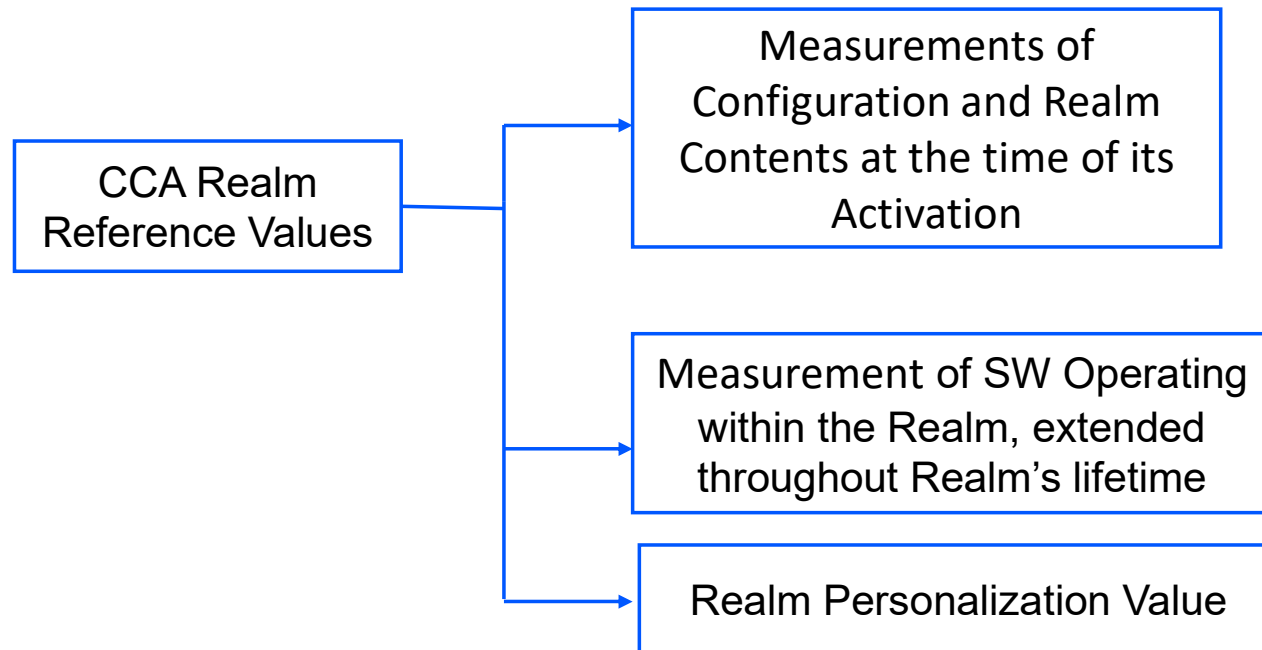
```
/ concise-mid-tag / {  
  / comid.tag-identity / 1 : {  
    / comid.tag-id / 0 : h'3f06af63a93c11e4979700505690773f'  
  },  
  / comid.triples / 4 : {  
    / comid.attest-key-triples / 3 : [[  
      / environment-map / {  
        / comid.class / 0 : {  
          / comid.class-id (implementation id) / 0 :  
            / tagged-bytes / 560(  
              h'61636d652d696d706c656d656e746174  
                696f6e2d69642d303030303030303031') },  
          / comid.instance / 1 :  
            / tagged-ueid-type (instance id) / 550(h'014ca3e4f50bf248c39787020d68ffd05c88767751bf2645ca923f57a98becd296' )},  
          [ / tagged-pkix-base64-key-type / 554(  
            "-----BEGIN PUBLIC KEY-----\MHYwEAYHKoZIzj0CAQYFK4EEACIDYgAEIShnxS4rlQiwpcCpBWDzINLfqiG911FP\\  
n8akBr+fh94uxHU5m+Kijivp2r2oxxN6MhM4tr8mWQli1P61xh3T0ViDREbF26DGOEYfbAjWjGNN7pZf+6A4OTHYqEryz6m7U\n-----END PUBLIC  
KEY-----\n" )]   ]]]}
```

arm

Arm CCA Realm Endorsements

Realm Reference Values

Arm CCA allows the initial & run time state of a Realm to be Attested



corim_profile= "tag:arm.com,2025:endorsements/cca_realm#1.0.0"

Realm Reference Values - RIM

- Realm Initial Measurements
 - Carried in reference-triple-record
 - Expressed using CoRIM Measurement Map
 - Uses a fixed Mkey – “cca.rim” to uniquely identify the measurement
 - Comid.Digests (key 2) carries the measurements

```
/ measurement-map (RIM) / {  
  / comid.mkey / 0 : "cca.rim",  
  / comid.mval / 1 : {  
    / comid.digests / 2 : [  
      "sha-256",  
  
      h'311314ab73620350cf758834ae5c65d9  
  
      e8c2dc7febe6e7d9654bbe864e300d49'  
    ]}  
  }
```

Realm Reference Values - REM

- Realm Extended Measurements
 - Expressed using CoRIM Measurement Map
 - This profile can contain a fixed 4 Extended Measurements
 - Mkey ("cca.rem0".."cca.rem3") uniquely identify each measurement

```
/ measurement-map (REM[0]) / {  
  / comid.mkey / 0 : "cca.rem0",  
  / comid.mval / 1 : {  
    / comid.digests / 2 : [  
      "sha-256",  
      h'24d5b0a296cc05cbd8068c5067c5bd47  
      3b770dda6ae082fe3ba30abe3f9a6ab1'  
    ]},  
  ]},
```

;; Similarly for REM 1,2 & 3

Realm Personalization Value

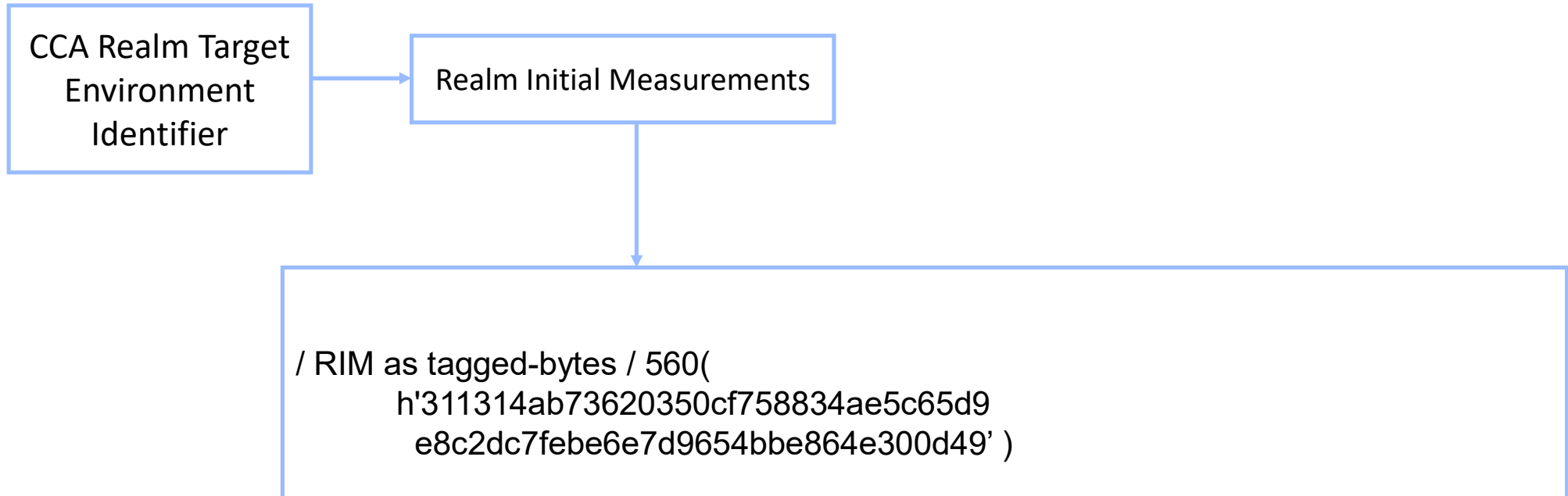
- Provided by Host
 - To distinguish between Realms that have the same Initial Measurements
 - Different behavior
 - RPV is encoded using a raw-value (key 4) , in a measurement values map
 - Mkey of “cca.rpv” is used to uniquely identify it inside the list of measurements

```
/ measurement-map (RPV) / {  
  / comid.mkey / 0 : "cca.rpv",  
  / comid.mval / 1 : {  
    / comid.raw-value / 4 : 560(  
      h'54686520717569636b2062726f776e20  
      666f78206a756d7073206f7665722031  
      33206c617a7920646f67732e54686520  
      717569636b2062726f776e20666f7820') }}  
}
```

Realm Endorsements

- Realm Attestation Verification Key Endorsements are NOT needed as
 - The key material needed to verify the Realm Evidence is inline in the CCA Evidence Token

Realm Environment Identifiers



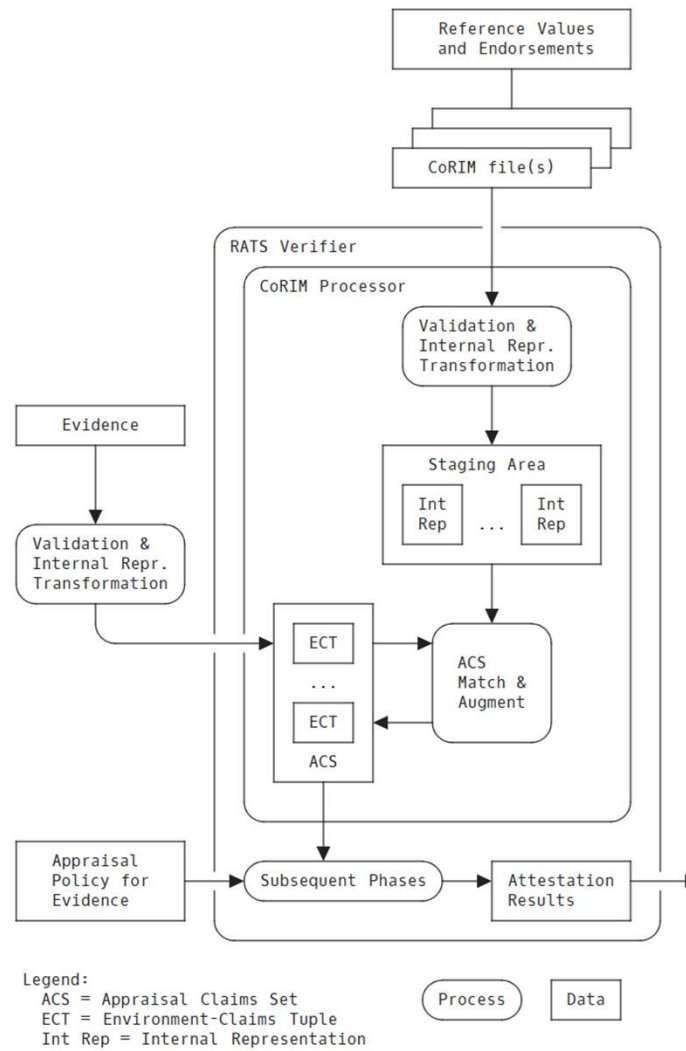
Realm Environment Identifiers

- Realms do not have an explicit Class or Instance Identifiers
- Realm Initial Measurements (RIM) is unique and stable to serve as an Identifier for the Realm Target Environment
- Realm Personalization Value (RPV) is used when Multiple Realms are instantiated with identical Realm Initial Measurements, but different behavior

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CoRIM based Appraisal

CoRIM Processor



Reference Section 8 of CoRIM Draft

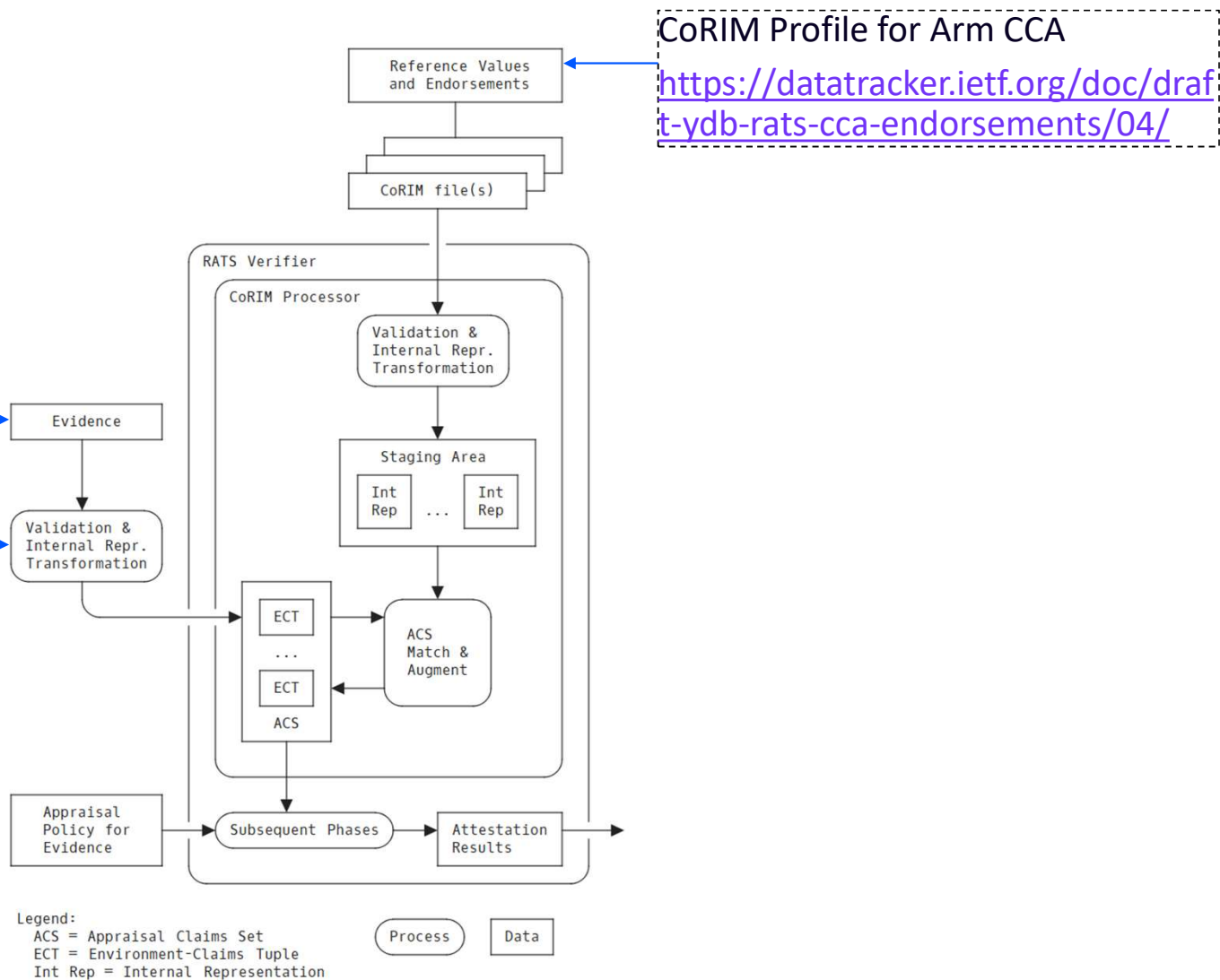
Mapping Arm CCA

Arm CCA Attestation token Draft
<https://datatracker.ietf.org/doc/draft-ffm-rats-cca-token/04/>

CoRIM Profile for Arm CCA
<https://datatracker.ietf.org/doc/draft-ydb-rats-cca-endorsements/04/>

Section – 3.1.5

Section– 3.2.4



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Merci

Danke

Gracias

Grazie

谢谢

ありがとう

Asante

Thank you

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